

Assignment 3A

Global Baseline Estimate

Approach

I'm trying to predict the ratings that are missing based on two things: how generous each critic usually is with their ratings and how well-liked each movie is overall. I'll start by loading the ratings exactly as they were collected, keeping the missing values blank instead of filling anything in right away. From there, I'll calculate three averages: the overall average rating, the average rating for each movie, and the average rating for each critic. I can then see how far above or below the overall average each movie and critic tends to be. For a missing rating, I'll add those two differences to the overall average to get a predicted rating. Since the calculation could technically give me a number below 1 or above 5, I'll make sure the final predictions stay within that range. Before applying it to the whole dataset, I'll also work through one prediction by hand and compare it with the example in the spreadsheet to make sure the code is doing what it's supposed to.

There are a few obstacles I will come across such as: some critics only rated one or two movies, so their average isn't based on much information. That makes predictions involving those critics less reliable than predictions for critics who rated a lot of movies. The same issue can come up with movies that only have a few ratings. Another thing is that the data needs to be loaded from a URL rather than from a file already saved on my computer, so I'll need to download it to a temporary file in the code first.

Code Deliverable

```
library(readxl)
```

```
Warning: package 'readxl' was built under R version 4.6.1
```

```
library(knitr)
```

```
raw <- read_excel("MovieRatings.xlsx")
critics <- raw$Critic
ratings <- as.matrix(raw[, -1])
rownames(ratings) <- critics
storage.mode(ratings) <- "numeric"
kable(ratings, caption = "Raw critic x movie ratings (blank = not rated)")
```

	CaptainAmer- ica	Dead- pool	Frozen	Jungle- Book	PitchPerfect2	StarWars- Force
Burton	NA	NA	NA	4	NA	4
Charley	4	5	4	3	2	3
Dan	NA	5	NA	NA	NA	5
Dieudonne	5	4	NA	NA	NA	5
Matt	4	NA	2	NA	2	5
Mauricio	4	NA	3	3	4	NA
Max	4	4	4	2	2	4
Nathan	NA	NA	NA	NA	NA	4
Param	4	4	1	NA	NA	5
Parshu	4	3	5	5	2	3
Prashanth	5	5	5	5	NA	4
Shipra	NA	NA	4	5	NA	3
Sreejaya	5	5	5	4	4	5
Steve	4	NA	NA	NA	NA	4
Vuthy	4	5	3	3	3	NA
Xingjia	NA	NA	5	5	NA	NA

The global baseline estimate for any (user, movie) pair is $\hat{r}_{u,i} = \mu + b_i + b_u$, where μ is the raw overall mean, b_i is the movie's average deviation from μ , and b_u is the user's average deviation from μ .

```
mu <- mean(ratings, na.rm = TRUE)

movie_avg <- colMeans(ratings, na.rm = TRUE)
movie_bias <- movie_avg - mu

user_avg <- rowMeans(ratings, na.rm = TRUE)
user_bias <- user_avg - mu

kable(data.frame(Movie = names(movie_bias), Bias = round(movie_bias, 3)))
```

	Movie	Bias
CaptainAmerica	CaptainAmerica	0.338
Deadpool	Deadpool	0.510

	Movie	Bias
Frozen	Frozen	-0.207
JungleBook	JungleBook	-0.034
PitchPerfect2	PitchPerfect2	-1.220
StarWarsForce	StarWarsForce	0.219

```
kable(data.frame(Critic = names(user_bias), Bias = round(user_bias, 3)))
```

	Critic	Bias
Burton	Burton	0.066
Charley	Charley	-0.434
Dan	Dan	1.066
Dieudonne	Dieudonne	0.732
Matt	Matt	-0.684
Mauricio	Mauricio	-0.434
Max	Max	-0.601
Nathan	Nathan	0.066
Param	Param	-0.434
Parshu	Parshu	-0.268
Prashanth	Prashanth	0.866
Shipra	Shipra	0.066
Sreejaya	Sreejaya	0.732
Steve	Steve	0.066
Vuthy	Vuthy	-0.334
Xingjia	Xingjia	1.066

Predictions are put in a vector as an outer sum of the two bias vectors plus μ , then put it on the scale.

```
pred <- outer(user_bias, movie_bias, "+") + mu
pred <- pmin(pmax(pred, 1), 5)
kable(round(pred, 1))
```

	CaptainAmer- ica	Dead- pool	Frozen	Jungle- Book	PitchPerfect2	StarWars- Force
Burton	4.3	4.5	3.8	4.0	2.8	4.2
Charley	3.8	4.0	3.3	3.5	2.3	3.7
Dan	5.0	5.0	4.8	5.0	3.8	5.0
Dieudonne	5.0	5.0	4.5	4.6	3.4	4.9
Matt	3.6	3.8	3.0	3.2	2.0	3.5
Mauricio	3.8	4.0	3.3	3.5	2.3	3.7
Max	3.7	3.8	3.1	3.3	2.1	3.6
Nathan	4.3	4.5	3.8	4.0	2.8	4.2
Param	3.8	4.0	3.3	3.5	2.3	3.7
Parshu	4.0	4.2	3.5	3.6	2.4	3.9
Prashanth	5.0	5.0	4.6	4.8	3.6	5.0
Shipra	4.3	4.5	3.8	4.0	2.8	4.2
Sreejaya	5.0	5.0	4.5	4.6	3.4	4.9
Steve	4.3	4.5	3.8	4.0	2.8	4.2
Vuthy	3.9	4.1	3.4	3.6	2.4	3.8
Xingjia	5.0	5.0	4.8	5.0	3.8	5.0

```
pred["Param", "PitchPerfect2"]
```

```
[1] 2.279859
```

```
index <- which(!is.na(ratings), arr.ind = TRUE)
act <- ratings[index]
predic <- pred[index]

rmse <- sqrt(mean((act - predic)^2))
rmse
```

```
[1] 0.7393273
```

Not every value in the output of the above code matches what is in the excel sheet. This is expected as there is never a model that is 100%. The Root Mean Square Error is approximately 74%.

```

rec <- function(critic) {
  if (!(critic %in% rownames(ratings))) {
    return(paste("Critic not found"))
  }
  u <- names(which(is.na(ratings[critic, ])))
  if (length(u) == 0) {
    return(paste("Already rated every movie in the dataset."))
  }

  scores <- predicted[critic, u]
  movie <- names(sort(scores, decreasing = TRUE))[1]
  score <- round(sort(scores, decreasing = TRUE)[1], 2)

  paste0("Recommended: ", movie,
        " (predicted rating: ", score, ")")
}

name <- readline(prompt = "Enter name: ")

```

```
Enter name:
```

```
cat(rec(name), "\n")
```

```
Critic not found
```

When you run the above code, you will be prompted to provide a name and it will return a movie recommendation for the person based on the estimate.

Conclusion

The Global Baseline Estimate fills in all of the missing ratings on the basis of just two factors: how generous a particular critic usually is and how popular a particular movie is. The score of RMSE shows the level of proximity of such simple prediction to the actual ratings made by the critics and also how big of a gap remains between them. Some critics reviewed only one or two films, so the generosity factor for them is estimated on the basis of very small amounts of data, thus making such prediction rather approximate.

There are several ways of improving the results obtained. First of all, the transition to a personalized approach is possible; it would mean finding similar users and movies for each critic and then comparing them in order to predict the ratings for critics similar enough. In addition, it would be useful to obtain more data from the critics who provided not many reviews in order to have more data points.